

Marking Scheme: CS 301 Midterm

Exam: Midterm — Data Structures & Algorithms

Course: CS 301

Pages per student: 2

Question 1 (10 marks)

Explain the time complexity of QuickSort in average and worst cases. Derive the recurrence.

Criteria:

- Correct average case $O(n \log n)$ [3 marks]
- Worst case $O(n^2)$ mentioned [3 marks]
- Partition step explained as $O(n)$ [2 marks]
- Recurrence $T(n)=2T(n/2)+O(n)$ stated [2 marks]

Acceptable patterns: $O(n \log n)$, $n \log n$, $n \log n$, $O(n^2)$, $O(n^2)$

Question 2 (10 marks)

Compare BFS and DFS. When would you use each?

Criteria:

- BFS uses a queue; DFS uses a stack [2 marks]
- BFS finds shortest path in unweighted graphs [3 marks]
- DFS useful for topological sort / cycle detection [3 marks]
- Correct time complexity $O(V+E)$ for both [2 marks]

Acceptable patterns: queue, stack, $O(V+E)$, shortest path, topological